

Nuru Pathway — Backend & App Audit

Scope: what the native iOS member app (`nuru-member-ios`) uses, what works, what's broken, what to add

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Sources: backend monorepo `pathway` (`packages/backend/src/modules/*`, OpenAPI `packages/shared/src/openapi/openapi.yaml`), iOS client `nuru-member-ios` (`NuruMember/Networking/MemberAPI.swift` + direct `APIClient` calls), live local stack (`http://localhost:8080/v1`, Postgres `nuru@localhost:5432/nuru`)

Method: static route extraction from all 27 backend modules; cross-reference against every API call in the iOS app; live curl of ~40 GET endpoints with a `student1@dev.local` dev token; read-only SQL row counts; code-level verification of each claimed gap. No code or data was modified.

1. Executive summary

The backend exposes roughly **300 routes**; about **190 are admin/portal routes** and about **110 are member-facing**. The iOS app wires ~**85 of the member-facing routes** through `MemberAPI.swift` (plus auth/refresh in `APIClient/AuthStore`, sync in `OfflineStore`, and a handful of direct calls in `ProfileView/GivingView`). Live verification against the local stack shows the wired surface is in **very good health**: home dashboard, pathway/modules/quizzes, growth (devotional, memory verses, plans, resources, mentor), prayer journal & wall, chat inbox/send, calendar/events/RSVP, giving history/intents/schedules, gifts assessment, badges, notifications, and offline sync all return correct, well-shaped data.

The problems cluster in four places:

- 1. Dangling contracts** — GET `/certificates` returns a `download_url` of `/media/certificates/{code}` that **no route serves** (confirmed 404); the chat conversation list omits the DM peer's `user_id`, forcing the iOS mentor screen into a name-matching hack.
- 2. Backend-ready, iOS-unwired** — space join (POST `/chat/spaces/:id/join`), DM creation (POST `/chat/dms`), QR event check-in (POST `/events/:id/attendance`), event posts/"buzz" (GET/POST `/events/:id/posts`), avatar upload, push-token registration (POST `/me/devices`), per-pillar score details, level exams, community threads, share-prayer-to-wall — all exist and are documented in OpenAPI but the app never calls them.
- 3. Provider keys absent** — assistant (`GROQ_API_KEY`/`GEMINI_API_KEY`) and scripture (`YOUVERSION_APP_KEY`) return `UPSTREAM_UNAVAILABLE` locally; both must be configured in prod.
- 4. Genuinely missing** — a member notification-preferences endpoint (prefs are written once at onboarding, and the iOS onboarding flow doesn't exist, so `notification_preferences` has **0 rows**); a live-stream/radio source for the Home "Radio" button and LiveNow card; a plan-day reflection endpoint; a PayPal capture call in the iOS flow (opens the approval URL but never calls POST `/giving/paypal/capture`, so PayPal gifts can never settle from the app).

2. Member-facing route inventory

Legend — **WIRED**: called from the iOS app (`MemberAPI.swift` or direct `APIClient` call). **UNUSED**: route exists, app never calls it. **MISSING**: the app/design needs it and no route exists (listed in §2.15). Admin/portal routes (`/admin/*`, leader-only engagement/assessment-review routes) are out of scope and counted only in the appendix.

2.1 Identity & auth (`modules/identity`)

Method	Path	Purpose	Status
POST	<code>/auth/login</code>	Email+password login (may return 2FA challenge)	WIRED
POST	<code>/auth/login/mfa</code>	Complete a 2FA login challenge	WIRED
POST	<code>/auth/register</code>	Create an account and sign in	WIRED
POST	<code>/auth/token/refresh</code>	Rotate the refresh token	WIRED

Method	Path	Purpose	Status
POST	/auth/password/forgot	Request reset (dev returns <code>dev_token</code>)	WIRED
POST	/auth/password/reset	Set new password from token	WIRED
POST	/auth/mfa/enroll	Begin TOTP enrollment	WIRED
POST	/auth/mfa/verify	Confirm enrollment code	WIRED
POST	/auth/mfa/disable	Turn off 2FA	WIRED
POST	/auth/logout	Revoke the session server-side	UNUSED
POST	/auth/oauth/:provider	KingsChat/OIDC social login	UNUSED
POST	/auth/dev-login	Dev-only instant login	N/A (dev)
GET	/me	Profile + enrollment summary	WIRED
PATCH	/me	Update profile fields	WIRED
POST	/me/password	Change password	WIRED
GET	/me/activity	Member's own audit/activity feed	UNUSED
POST	/me/devices	Register a push token (APNs/FCM)	UNUSED
POST	/me/onboarding	Legacy onboarding marker	UNUSED

2.2 Home dashboard (`modules/home`, plus member routes in `adminops`)

Method	Path	Purpose	Status
GET	/me/home/greeting	Warm daily greeting line	WIRED
GET	/me/home/next-action	Next-best-action hero card	WIRED
GET	/me/home/verse	Tailored verse for today	WIRED
GET	/me/home/verse/reactions	Community reaction counts	WIRED
POST	/me/home/verse/reactions	Set my reaction	WIRED
GET	/me/rhythm/today	Today's three daily rhythms	WIRED
POST	/me/rhythm/complete	Mark a rhythm done (idempotent)	WIRED
GET	/home/featured-cell	"This week at Nuru" cell card	WIRED

2.3 Curriculum & pathway (`modules/curriculum`, `progress`, `assessment`)

Method	Path	Purpose	Status
GET	/me/pathway	Level trail with progress + status	WIRED
GET	/levels	Level catalogue	UNUSED
GET	/levels/:n/modules	Module trail for a level	WIRED
GET	/modules/:id	Full lesson (server-gated, \$1.9)	WIRED
GET	/scripture?ref=	Passage text (YouVersion-backed)	WIRED
POST	/modules/:id/complete	Finish a non-quiz module (+reflection)	WIRED
GET	/modules/:id/quiz	Server-assembled quiz	WIRED
POST	/modules/:id/quiz/attempts	Submit answers, scored server-side	WIRED
GET	/modules/:id/reflection	My reflection + review status	UNUSED

Method	Path	Purpose	Status
GET	/levels/:n/exam	Level exam paper	UNUSED
POST	/levels/:n/exam/attempts	Submit level exam	UNUSED
POST	/levels/:n/reflection	Level-gate reflection submission	UNUSED
POST	/events/:id/attendance	QR/manual event check-in (member)	UNUSED

2.4 Growth content (modules/growth-content)

Method	Path	Purpose	Status
GET	/growth/devotional	Today's devotional (+my reflection)	WIRED
POST	/growth/devotional/reflection	Save reflection; marks rhythm	WIRED
GET	/growth/memory-verses	Memory-verse set with status	WIRED
POST	/growth/memory-verses/practice	Log a practice attempt	WIRED
GET	/growth/plans	Reading-plan catalogue	WIRED
GET	/growth/plans/:id	Plan with day-by-day breakdown	WIRED
POST	/growth/plans/:id/start	Enrol in a plan	WIRED
POST	/growth/plans/:id/complete-day	Mark a whole day done	WIRED
POST	/growth/segments/:id/complete	Mark one plan-day segment done	WIRED
GET	/growth/resources	Library (books/audio/video/articles)	WIRED
GET	/growth/mentor	Assigned discipler + meeting notes	WIRED
GET	/home/disciplers	"Meet your discipler" carousel	WIRED

2.5 Personal growth data (modules/growth)

Method	Path	Purpose	Status
GET	/me/prayers	Private prayer journal	WIRED
PUT	/me/prayers	Upsert entry (idempotent)	WIRED
DELETE	/me/prayers/:id	Delete entry	WIRED
GET	/me/verses	Saved-verse library	WIRED
PUT	/me/verses	Save/update a verse	WIRED
DELETE	/me/verses/:id	Remove a saved verse	WIRED
GET	/me/gifts	Spiritual-gifts profile	WIRED
GET	/gifts/questions	Likert question set	WIRED
POST	/gifts/assessments	Submit answers → new profile	WIRED

2.6 Scores (modules/scores)

Method	Path	Purpose	Status
GET	/me/scores	Five growth scores + weighted overall	WIRED
GET	/me/scores/habits	Habits pillar detail	UNUSED
GET	/me/scores/curriculum	Curriculum pillar detail	UNUSED

Method	Path	Purpose	Status
GET	/me/scores/attendance	Attendance pillar detail	UNUSED
GET	/me/scores/prayer	Prayer pillar detail	UNUSED
GET	/me/scores/word	Word pillar detail	UNUSED

2.7 Calendar, events & moments (modules/calendar)

Method	Path	Purpose	Status
GET	/calendar?from=&to=	Occurrences in a date window	WIRED
GET	/calendar/series	Followable event series	WIRED
POST	/calendar/series/:id/follow	Toggle following a series	WIRED
GET	/home/featured-event	Admin-featured event	WIRED
GET	/events/:id	Full event detail (occurrence id)	WIRED
POST	/events/:id/rsvp	Set RSVP	WIRED
GET	/me/rsvps	My RSVPs	WIRED
GET	/me/cell-summary	My cell card	WIRED
GET	/moments	Curated photo gallery	WIRED
GET	/events/:id/posts	Event buzz/posts feed	UNUSED
POST	/events/:id/posts	Post to an event's buzz	UNUSED
POST	/events/:id/posts/:postId/react	React to a buzz post	UNUSED
POST	/calendar/parse	Natural-language event parse	UNUSED

2.8 Announcements (modules/announcements)

Method	Path	Purpose	Status
GET	/home/featured-announcement	Admin-featured announcement	WIRED
GET	/me/announcements	My announcements	WIRED
GET	/announcements/:id	Full announcement (carousel + body)	WIRED
POST	/announcements/:id/open	Mark opened (best-effort)	WIRED

2.9 Giving & products (modules/financial) — online-only, \$5.6

Method	Path	Purpose	Status
GET	/giving/history	Gift history	WIRED
POST	/giving/intents	Create a gift intent (M-Pesa/Airtel/PayPal)	WIRED
GET	/giving/transactions/:id	Gift detail incl. ledger trail	WIRED
GET	/giving/schedules	Recurring gifts	WIRED
POST	/giving/schedules	Create a recurring gift	WIRED
POST	/giving/schedules/:id/cancel	Cancel a schedule	WIRED

Method	Path	Purpose	Status
POST	/giving/paypal/capture	Capture an approved PayPal order	UNUSED ■ (breaks PayPal settle — see §4.6)
GET	/giving/statement.pdf	Annual statement PDF	UNUSED (statement screen renders natively)
GET	/giving/transactions/:id/receipt.pdf	Receipt PDF	UNUSED
GET	/products	Purchasable products	UNUSED (0 rows)
POST	/products/:id/purchase	Buy a product	UNUSED

2.10 Chat ([modules/chat](#))

Method	Path	Purpose	Status
GET	/chat/conversations?scope=mine	Inbox (Spaces/DMs/Groups + discover_spaces)	WIRED
GET	/chat/conversations/:id	Thread with messages	WIRED
POST	/chat/conversations/:id/messages	Send a message (idempotent)	WIRED
POST	/chat/conversations/:id/read	Mark thread read	WIRED
POST	/chat/messages/:id/reactions	Toggle emoji reaction	WIRED
POST	/chat/spaces/:id/join	Join a public space	UNUSED ■ (backend ready; iOS "Browse spaces" is a dead button)
POST	/chat/dms	Create/get a DM by user_id	UNUSED ■ (would fix the mentor name-hack)
GET	/chat/people	People directory for new DMs	UNUSED
POST	/chat/attachments/sign	Cloudinary signed-upload for attachments	UNUSED
PATCH / DELETE	/chat/messages/:id	Edit / delete own message	UNUSED
GET	/chat/messages/:id/readers	Read receipts	UNUSED
POST	/chat/messages/:id/flag · /unflag	Report a message	UNUSED
POST	/chat/cells/:id/conversation, /chat/spaces	Leader-only creation	N/A (leader)

2.11 Prayer wall & community ([modules/prayer-wall](#), [community](#))

Method	Path	Purpose	Status
GET	/prayer-wall?sort=	Shared prayer requests	WIRED
GET	/prayer-wall/:id	One request + comments	WIRED
POST	/prayer-wall	Share a request	WIRED
POST	/prayer-wall/:id/reactions	Toggle ■ etc.	WIRED
POST	/prayer-wall/:id/comments	Encourage the requester	WIRED
POST	/prayer-wall/:id/answered	Author marks (un)answered	WIRED
DELETE	/prayer-wall/:id	Author removes request	WIRED
GET	/home/prayer-wall	Home carousel posts	WIRED
POST	/me/prayers/:id/share-to-wall	Promote a journal entry to the wall	UNUSED

Method	Path	Purpose	Status
GET/POST	/community/threads, /:id, /:id/comments	Cell discussion board	UNUSED (0 rows; iOS Community tab shows it as "coming soon")

2.12 Gamification, certificates, encouragements

Method	Path	Purpose	Status
GET	/me/achievements	Badges + streak	WIRED
GET	/badges	Badge catalogue	WIRED
GET	/members/:id/achievements	Another member's badges	UNUSED
GET	/cells/:id/milestones	Cell milestones	UNUSED
GET	/certificates	My certificates (with <code>download_url</code>)	WIRED
GET	/verify/:code	Public certificate verification	WIRED
GET	/levels/:n/encouragements	Level-gate encouragement messages	UNUSED (0 rows)

2.13 Notifications, media, assistant, proximity, sync, activity

Method	Path	Purpose	Status
GET	/me/notifications	Notification center + unread count	WIRED
POST	/me/notifications/read	Mark some/all read	WIRED
GET	/home/welcome-video	Homepage welcome video	WIRED
POST	/media/:id/reactions	Toggle media reaction	WIRED
POST	/me/avatar	Upload profile photo	UNUSED
GET	/media/:id/manifest, /media/url	Video manifest / URL broker	UNUSED
GET	/assistant/history	Prior assistant turns	WIRED
POST	/assistant/chat	Nuru AI companion reply	WIRED (needs provider key)
POST	/me/location	Share coarse geohash	WIRED
DELETE	/me/location	Opt out of location	WIRED
POST	/sync/push	Replay offline mutation queue	WIRED
POST	/sync/pull	Cursor-driven delta pull	WIRED
POST	/me/activity/screens	Screen-view analytics batch	UNUSED (0 rows)

2.14 Onboarding (`modules/onboarding`) — entire module unwired

Method	Path	Purpose	Status
GET	/onboarding	Step-machine state	UNUSED
PUT	/onboarding/steps/{profile,cell_selection,literacy_quiz,notifications,guardian_consent}	Save each step	UNUSED
GET	/onboarding/literacy-quiz	Literacy quiz paper	UNUSED
POST	/onboarding/finalize	Complete onboarding	UNUSED

Method	Path	Purpose	Status
GET	/directory/cell-groups	Cell picker directory	UNUSED

The native app registers via `POST /auth/register` and lands straight on Home; the onboarding flow (and with it the **only** write path to `notification_preferences`) is never exercised.

2.15 MISSING — no route exists

Needed by	What's missing		
Certificates screen	<code>GET /media/certificates/:code</code> — the URL every certificate's <code>download_url</code> points at (<code>certificates/service.ts:99</code>). Confirmed live: returns an Express plain-404 (route absent). PDFs are rendered and stored in the ObjectStore at issuance; they just cannot be fetched.		
Profile → notification toggles	Member <code>GET/PUT /me/notification-preferences</code> . The three iOS toggles are <code>@AppStorage</code> local-only (<code>ProfileView.swift:20-22</code>); the server table is written only by the unused onboarding step. <code>notification_preferences: 0 rows</code> .		
Home "Radio" button + Live Now card	Any live-stream source. Zero matches for `radio	<code>stream_url</code>	livestream` across all backend modules. The LiveNow card is driven purely by calendar occurrence timing; tapping "Radio" just opens the occurrence — there is nothing to play.
Reading-plan day screen	Plan-day reflection endpoint (e.g. <code>POST /growth/plans/:id/days/:n/reflection</code>). iOS comment confirms: <i>"no reflection binding for plan days"</i> (<code>ReadingPlansView.swift:641-642</code>).		
Mentor "Message" button	<code>peer_user_id</code> on each DM row in <code>GET /chat/conversations</code> . The SQL joins the peer (<code>od.user_id</code>) but only selects their name/avatar (<code>chat/service.ts:282-305</code>), so <code>MentorView.swift:17-28</code> matches the mentor's DM by full name . (Alternative fix needing no contract change: iOS calls <code>POST /chat/dms</code> with <code>mentor_user_id</code> , which <code>GET /growth/mentor</code> already returns.)		

3. Live verification — what works, what doesn't

3.1 Endpoint spot-checks (dev token, `student1@dev.local`)

Endpoint	Result
<code>GET /me</code>	■ Full profile (Ada Thriving, Student, cell + congregation ids)
<code>GET /me/pathway</code>	■ 6 levels, level 1 active 20/30 modules
<code>GET /me/rhythm/today</code>	■ <code>{prayer:false, word:false, reflection:false}</code>
<code>GET /me/home/next-action</code>	■ "Continue: God & His Nature" resume card
<code>GET /me/home/greeting · /verse · /verse/reactions</code>	■ All populated (verse from <code>daily_verses</code> , 365 rows seeded)
<code>GET /me/scores (+ /me/scores/word etc.)</code>	■ Overall 22 with all five pillars + component detail
<code>GET /me/achievements</code>	■ Shape correct; empty badges/streak (no <code>user_badges/user_streaks</code> rows)
<code>GET /me/notifications</code>	■ Works; empty (<code>notifications: 0 rows</code> — nothing has ever fanned out)

Endpoint	Result
GET /growth/devotional · memory-verses · plans · resources · mentor	■ All richly seeded (7 plans, 54 memory verses, 2 devotionals, 5 resources, mentor + 3 notes)
GET /home/featured-event · -announcement · -cell · welcome-video · discipplers · prayer-wall · /moments	■ All populated
GET /calendar · /calendar/series	■ Occurrences and 5 series
GET /events/{occurrence-id}	■ Works with occurrence-style ids (<code>seriesId:ISO-timestamp</code>); plain series ids 404 as expected
GET /events/{id}/posts	■ Route works; returns <code>{"data": []}</code> (<code>event_posts</code> : 0 rows)
GET /chat/conversations?scope=mine	■ 1 cell group conversation + <code>discover_spaces</code> array present
GET /prayer-wall	■ 5 posts
GET /giving/history · /schedules · statement.pdf	■ 7 transactions, valid PDF streamed
GET /me/prayers · /me/verses	■ Seeded incl. rows written via offline queue
GET /gifts/questions · /me/gifts	■ Question set served; profile empty (no assessments yet)
GET /certificates	■ Route fine but empty (<code>certificates</code> : 0 rows) — and see broken <code>download_url</code> §4.1
GET /badges	■ 6 badge definitions
GET /levels/1/modules vs /levels/3/modules	■ Level 1 full; level 3 returns <code>[]</code> (no published modules above level 2 — 71 modules total, seeded through level 2)
GET /levels/1/encouragements	■ Empty — table has 0 rows ; nobody has authored encouragements
GET /community/threads	■ Empty — feature dark on both ends
GET /scripture?ref=John 3:16	■ <code>UPSTREAM_UNAVAILABLE</code> — "Scripture is not configured" (<code>YOUVERSION_APP_KEY</code> unset; <code>scripture.ts:55</code>)
POST /assistant/chat	■ (verified via code, not curled — it writes history) Same pattern: provider selected only if <code>GROQ_API_KEY</code> or <code>GEMINI_API_KEY</code> set (<code>assistant/provider.ts:155-156</code>); otherwise <code>UPSTREAM_UNAVAILABLE</code> . <code>assistant_messages</code> : 0 rows.
GET /media/certificates/SOMECODE	■ Plain Express 404 — route does not exist

3.2 Database health (row counts, local dev DB)

Populated (healthy): `users` 10 · `levels` 6 · `modules` 71 · `module_progress` 51 · `attendance_logs` 23 · `event_series` 5 · `events` 9 · `event_moments` 6 · `memory_verses` 54 · `reading_plans` 7 · `devotionals` 2 · `daily_verses` 365 · `prayer_wall_posts` 5 · `prayer_entries` 2 · `saved_verses` 1 · `transactions` 7 · `ledger_entries` 12 · `announcements` 3 · `badges` 6 · `resources` 5 · `mentor_notes` 3 · `gift_question_sets` 1 · `chat_conversations/chat_messages` 1/1 · `media_assets` 1 · `member_location` 1 · `verse_reactions` 1.

Empty (features never exercised or never fed):

Table	Why it's empty	Signal
<code>notification_preferences</code>	Only write path is the unwired onboarding step	Prefs feature effectively dead
<code>notifications</code> , <code>notification_states</code> , <code>push_tokens</code>	No fan-out has run; iOS never registers a push token	No push pipeline end-to-end
<code>event_posts</code> , <code>event_post_reactions</code>	Buzz routes exist, no client posts	Feature dark
<code>event_rsvps</code> , <code>event_guests</code>	No RSVPs made yet (route works)	Data-only

Table	Why it's empty	Signal
discussion_threads, discussion_comments	Community threads unwired in iOS	Feature dark
certificates	No member has passed a level exam (exams also unwired in iOS)	Blocked chain: exam → cert → download
user_badges, user_streaks, gamification_events	No awards triggered yet	Data-only
assistant_messages	Assistant blocked on provider key locally	Config
level_encouragements	No admin has authored any	Content gap
devotional_reflections, reading_plan_progress, quiz_attempts, gift_assessments, video_progress, app_screen_events, giving_schedules, products, prayer_wall_comments	Awaiting normal usage / seeding	Data-only

4. Known gaps — verified against code

Each claim from the brief was checked against source; two turn out to be *client-side* gaps rather than missing backend routes.

4.1 Certificate download_url → **CONFIRMED backend bug**

`certificates/service.ts:99` emits `download_url: /media/certificates/{verification_code}` for every certificate. No module registers that route (grepped all 27; live request 404s at the Express layer). The PDF is rendered at issuance (`service.ts:75-81`) into the ObjectStore (`certificates/{code}.pdf`) — in dev an `InMemoryObjectStore`, so a real deployment also needs the S3/Cloudinary implementation dropped in. **Fix: add the serving route (auth or signed) + a persistent ObjectStore in prod.**

4.2 Chat space join → **CORRECTED: backend route EXISTS; iOS is unwired**

`POST /chat/spaces/:id/join` exists (`chat/index.ts:110`) and `discover_spaces` is returned by the inbox and decoded by the iOS model (`Models/Chat.swift:36,73`). But the iOS "Browse spaces" compose action (`ChatView.swift:459`) has no destination and no join call anywhere in the app. **Fix is iOS-only.**

4.3 Notification preferences → **CONFIRMED (both sides)**

No member route reads/writes `notification_preferences`; the only writer is `PUT /onboarding/steps/notifications`, and iOS has no onboarding flow. iOS toggles are `@AppStorage`-local (`ProfileView.swift:20-22, 374-376`). Table: 0 rows. **Fix: add GET/PUT /me/notification-preferences + wire the toggles.** Related: `POST /me/devices` (push-token registration) exists but iOS never calls it, and `push_tokens` is empty — even with prefs, no push can be delivered.

4.4 Event check-in / QR → **CORRECTED: member route EXISTS; iOS is unwired**

`POST /events/:id/attendance` (progress module, `index.ts:43-52`) validates a scan token, is idempotent, and is also replayable offline via `sync/push` (`attendance:scan` mutation kind). Admin-side manual check-in also exists (`POST /admin/events/:id/checkins`). The iOS app has no scanner or check-in call. **Fix is iOS-only** (QR scanner screen + call, or offline enqueue).

4.5 Event posts / buzz → **CONFIRMED: routes exist, unused**

`GET/POST /events/:id/posts` and `POST /events/:id/posts/:postId/react` all exist and respond correctly live (`{"data": []}`; `event_posts` 0 rows). Documented in OpenAPI. Ready for the iOS wiring reportedly in flight.

4.6 PayPal capture → **NEW FINDING: iOS flow cannot settle PayPal gifts**

`GivingView.swift:722-780`: the app creates the intent, opens `approve_url`, then polls `GET /giving/transactions/:id` for up to ~60 s. It **never calls** `POST /giving/paypal/capture` (`financial/index.ts:39-47`), which is what settles the ledger

after approval. Unless a return-URL/webhook path captures server-side, every PayPal gift stalls in `processing`. **Fix: call capture on return-to-app (universal link or poll-then-capture).**

4.7 Per-pillar score details → CONFIRMED unused

All five `GET /me/scores/{pillar}` detail endpoints work (curled `word, habits`) with component/detail breakdowns; iOS only calls the summary. Opportunity: a "Why is my score X?" drill-down screen at near-zero backend cost.

4.8 `levels/:n/encouragements` → CONFIRMED unused + no content

Route works, iOS never calls it, and `level_encouragements` has 0 rows. Two gaps: wiring and authoring.

4.9 Community threads → CONFIRMED dark on both ends

Routes work; iOS Community tab lists "Cohort Discussions" as a dead (`live: false`) row (`CommunityView.swift:23-24`). Sync already supports `discussion_threads:create` / `discussion_comments:create` offline mutations, and pull domains include both tables — the plumbing is fully built.

4.10 Prayer share-to-wall → CONFIRMED unused

`POST /me/prayers/:id/share-to-wall` exists (prayer-wall module); no iOS call. The journal entry sheet has no "Share to wall" action.

4.11 Assistant provider key → CONFIRMED

`buildProvider` (`assistant/provider.ts:155-156`): Groq preferred, Gemini fallback, else every chat throws `UPSTREAM_UNAVAILABLE`. Both `GROQ_API_KEY` and `GEMINI_API_KEY` are optional in `config/env.ts:79-82`. Same class of issue as scripture (`YOUVERSION_APP_KEY`, confirmed failing live). **Set both keys in prod or the Nuru companion and in-app Bible text are dead.**

4.12 Live-stream / radio → CONFIRMED absent

No route, column, or env var relating to streaming anywhere in the backend. Home's Radio button and LiveNow card (`HomeView.swift:284-390`) navigate to a calendar occurrence — honest, but there is nothing to listen to.

4.13 Plan-day reflection → CONFIRMED absent

Growth-content module has only the devotional reflection endpoint. No per-plan-day reflection route; iOS explicitly omitted the mock's reflection textarea for this reason.

4.14 Mentor peer-user-id → CONFIRMED contract gap

Detailed in §2.15. The data exists on both sides; only the conversation-list contract hides it.

5. Recommendations — prioritized roadmap

Effort: $S \approx \leq \text{half-day}$ · $M \approx 1\text{--}2 \text{ days}$ · $L \approx 3+ \text{ days}$, per surface.

P1 — broken promises & prod blockers

#	Feature	Backend work	iOS work	DB work	Why it matters	Effort
1	Certificate download	Add <code>GET /media/certificates/:code</code> streaming from ObjectStore (auth: owner or valid code); wire S3/Cloudinary ObjectStore in prod	Add "Download PDF" button using existing <code>downloadUrl</code>	none (PDF keys already stored)	The app already shows the URL; a graduate's certificate is the pathway's crowning artifact and today it 404s	B:S · iOS:S

#	Feature	Backend work	iOS work	DB work	Why it matters	Effort
2	PayPal settle	(Optional) also capture server-side on PayPal webhook	Call <code>POST /giving/paypal/capture</code> on return-to-app; deep-link return URL	none	Money path is silently broken — gifts stall in <code>processing</code> , eroding trust in giving	B:S · iOS:M
3	Assistant + scripture keys in prod	Set <code>GROQ_API_KEY</code> (or <code>GEMINI_API_KEY</code>) and <code>YOUVERSION_APP_KEY</code> on the VPS; add readiness check/log warning when absent	none	none	Two flagship features (Nuru companion, in-app Bible text) hard-fail without them	B:S
4	Notification preferences + push registration	<code>GET/PUT /me/notification-preferences</code> (upsert; reuse onboarding step logic); OpenAPI	Bind the three Profile toggles to the API; call <code>POST /me/devices</code> with the APNs token after permission grant	none (table exists)	Server can't respect member choices it never receives; <code>push_tokens</code> empty means <code>zero</code> push reach today	B:S · iOS:M
5	Space join + mentor DM	Add <code>peer_user_id</code> to DM rows in <code>listConversations</code> (one <code>SELECT</code> column); OpenAPI	Browse-spaces sheet (list <code>discover_spaces</code> → <code>POST /chat/spaces/:id/join</code>); mentor Message button → <code>POST /chat/dms</code> with <code>mentor_user_id</code>	none	Chat discovery is a dead button; mentor messaging rests on a name-collision-prone hack	B:S · iOS:M

P2 — high-value wiring of ready backend

#	Feature	Backend work	iOS work	DB work	Why it matters	Effort
6	Event buzz (posts)	none — routes live	Posts feed + composer + reactions on event detail (reportedly in flight)	seed demo posts	Turns events from listings into community moments	iOS:M
7	QR event check-in	none — route + offline <code>attendance:scan</code> ready	Scanner screen; enqueue offline	none	Attendance feeds the attendance score pillar; members currently can't check in at all	iOS:M
8	Level exams + encouragements	none (routes ready); author <code>level_encouragements</code> content	Exam screen (<code>GET /levels/:n/exam</code> , submit attempts); show encouragements at level gates	seed encouragements	Level progression — the core pathway loop — cannot complete in the native app today; certificates depend on it	iOS:L · content:S
9	Score drill-downs	none — five detail endpoints live	"Why this score?" sheet per pillar using components/detail	none	Makes growth scores explainable and actionable instead of a bare number	iOS:M
10	Community threads	none — routes + offline sync ready	Cohort Discussions screen (list/create/comment)	seed a starter thread per cell	The Community tab advertises it; cell-level belonging is the retention engine	iOS:L
11	Notifications fan-out	Verify outbox → <code>notifications</code> fan-out jobs run in prod (table empty even for in-app rows); wire <code>PUSH_PROVIDER_KEY</code>	none beyond #4	none	Notification center is an empty room until something writes to it	B:M

P3 — new product surface

#	Feature	Backend work	iOS work	DB work	Why it matters	Effort
12	Live stream / radio	GET /live (or stream_url + is_live on occurrences/series); admin CRUD for stream config	Player sheet behind the Radio button + LiveNow CTA	live_streams table or columns on event_series	The Home UI already promises it; live services are the strongest weekly touchpoint for remote members	B:M · iOS:M · DB:S
13	Plan-day reflection	POST /growth/plans/:id/days/:n/reflection (+ GET with day payload); count toward reflection rhythm	Reflection textarea on the plan-day screen	reading_plan_day_reflections table	Reflection is one of the three daily rhythms; plans are where members read most	B:M · iOS:S · DB:S
14	Share prayer to wall	none — route exists	"Share to wall" action on journal entries	none	Bridges private discipline to communal encouragement in one tap	iOS:S
15	Avatar upload + logout	none — POST /me/avatar, POST /auth/logout exist	Photo picker on the profile pencil; call logout on sign-out	none	Identity/personal presence in chat, wall, disciplers; clean session hygiene	iOS:S
16	Chat depth (attachments, edit/delete, receipts, flag)	none — all routes exist	Attachment picker via chat/attachments/sign; long-press actions	none	Brings native chat to parity with the web portal and member expectations	iOS:L
17	Screen analytics	none — POST /me/activity/screens exists	Batch screen events from the router	none	app_screen_events is empty; engagement scoring and the admin Intelligence grid are flying blind on native usage	iOS:S

6. Appendix

6.1 Surface counts

- Backend routes total (all modules): ~300 (258 paths documented in OpenAPI — coverage spot-checks all passed, including the unused buzz and join routes)
- Admin/portal-only routes: ~190 across adminops, system, curriculum admin, financial admin, media admin, announcements admin, growth-content admin, certificates admin, gamification admin, proximity admin, encouragements admin, community moderation, assessment review, engagement (leader), calendar admin
- Member-facing routes: ~110 · wired from iOS: ~85 · unused: ~25 · missing: 5 (§2.15)

6.2 Offline sync contract (working)

- Push mutation kinds accepted: module_progress:complete, quiz_attempts:submit, level_exam_attempts:submit, interaction_events:record, video_progress:update, event_rsvps:set, attendance:scan, prayer_entries:upsert/delete, saved_verses:save/delete, discussion_threads:create, discussion_comments:create, chat_messages:create, chat_reactions:toggle, chat_reads:set, gift_assessments:submit. Money domains are rejected at the source on both client and server (§5.6).
- Pull domains mirrored in OfflineStore.pullIdField match the backend's PULL_DOMAINS; delta pull with tombstones verified working live (prayer/verse rows written via the queue appear in reads).

6.3 Notable environmental facts

- Dev login (POST /auth/dev-login) available; local stack healthy (/healthz, /readyz ok).
- Event detail ids are **occurrence ids** (seriesId:ISO-start) — the featured-event series_id alone will 404 on GET /events/:id; clients must pass the occurrence id (the iOS calendar path does this correctly).
- Levels 3–6 have no published modules yet (71 modules cover levels 1–2); GET /levels/3/modules correctly returns an empty list rather than leaking content (hard-lock §1.9 intact).

End of audit.